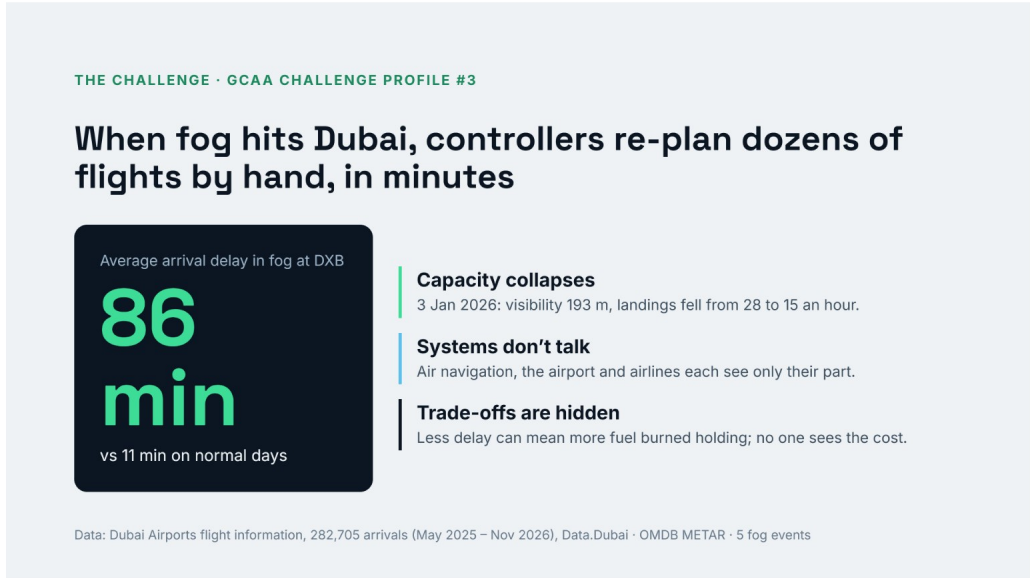
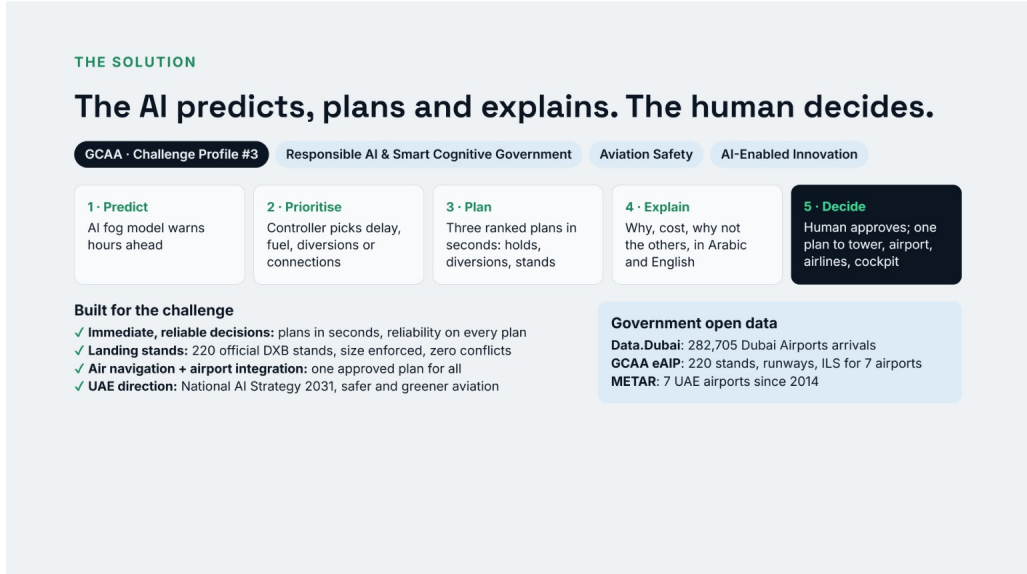


Opens after the 10-second video. Say: "That controller is not fiction. This is DABAB Copilot."



That morning was real. On 3 January, fog closed Dubai International. Across 18 months of Dubai Airports data, fog days average 86 minutes of delay, against 11 on normal days. Controllers decide who holds, who diverts and where, by hand, with three systems that don’t talk.



This is GCAA’s Challenge Profile 3, under Responsible AI, Aviation Safety and AI-enabled innovation. DABAB is a decision copilot for controllers. It warns before the fog, builds three recovery plans in seconds, explains each in Arabic and English, and a human always approves. It answers the challenge line by line, and it runs on government open data: Dubai Airports flights, the GCAA aeronautical publication and airport weather reports.

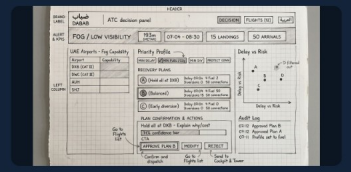
THE TEAM

Two builders, one working system on real data

Ali Eissa
Engine and AI: data, fog model, optimizer, explanations, backend

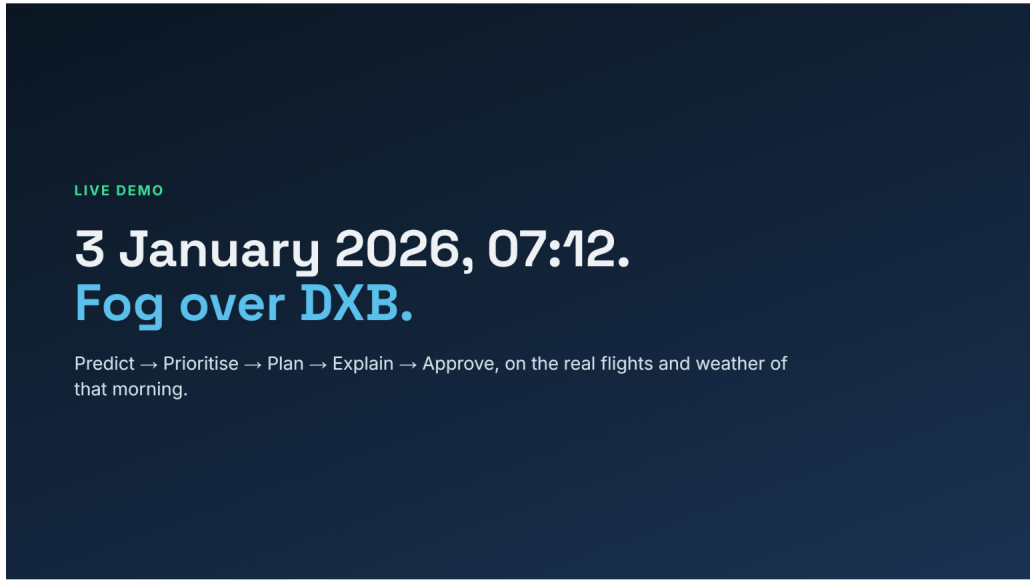
Mohammed Hishaam
Product and pitch: ATC panel, user journey, business model

> *"Fog is the UAE's most disruptive weather for aviation, and the decision is still made by hand."*

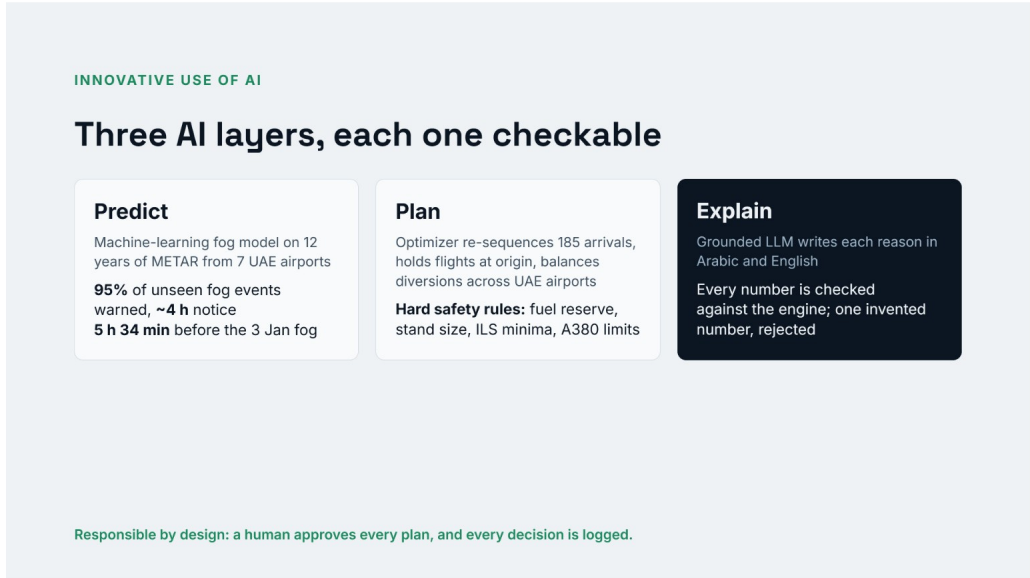


Wireframe → working panel with live backend
282,705 arrivals · 220 GCAA stands · fog model on 12 years of METAR

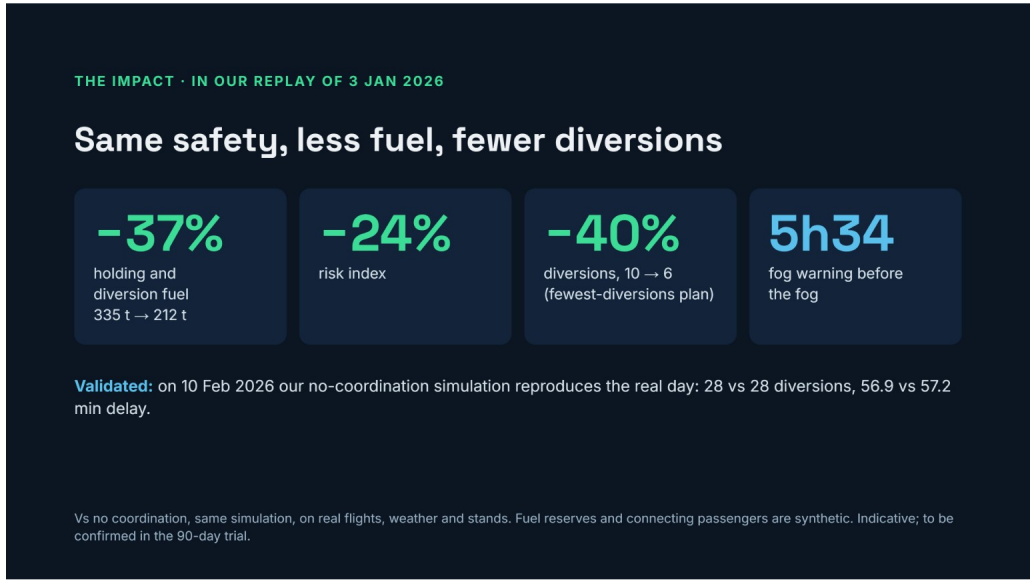
We are two people. In this hackathon we went from a paper wireframe to a working system on real data. (Switch to the live dashboard.)



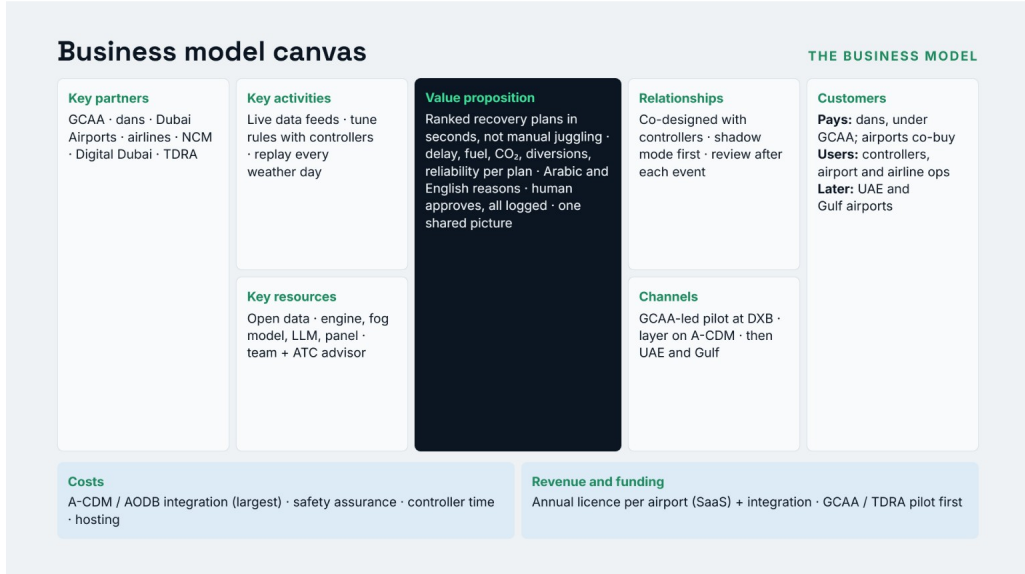
Switch to the dashboard (backup: screen recording). 1) Header: the AI fog model raised a Watch at 01:30, five and a half hours before fog hit at 07:04; on the right, the real weather reports. 2) Click Minimum fuel: the controller chooses the priority; safety rules never change. 3) Plans: three plans with delay, diversions, fuel, connections and reliability. In our replay, against no coordination, the recommended plan cuts holding and diversion fuel by 37% and the risk index by 24%. 4) Click plan C: why this plan, what it costs, why not the others, written by the AI using only the engine’s numbers. 5) Approve: one click sends the plan to tower, airport stands, airline ops and the cockpit, and it is logged. The human decides.



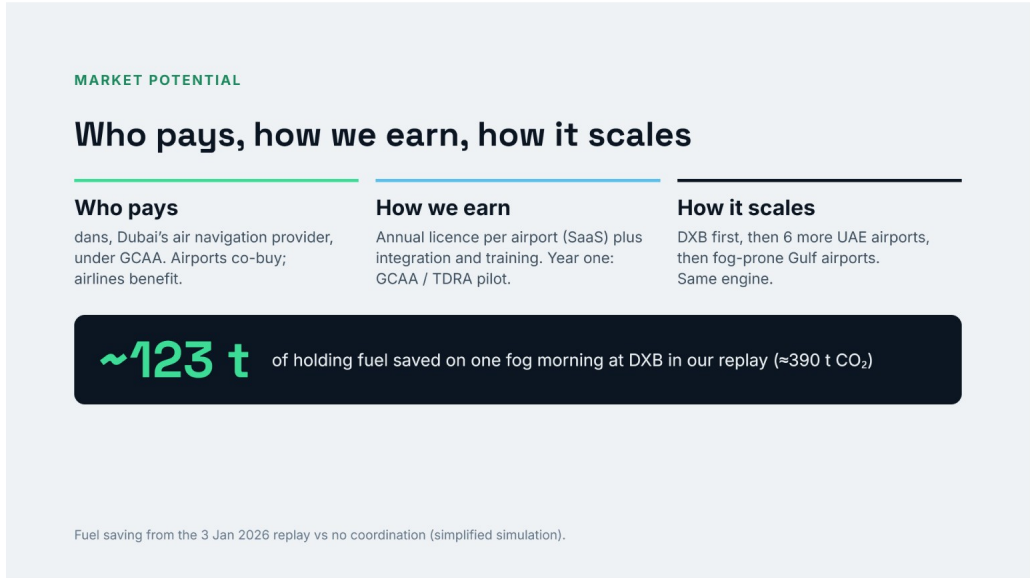
Three AI layers: a fog prediction model trained on twelve years of airport weather, which warned of 95% of fog events it had never seen; an optimizer with hard safety rules; and a language model that explains, but can only quote numbers the engine produced.



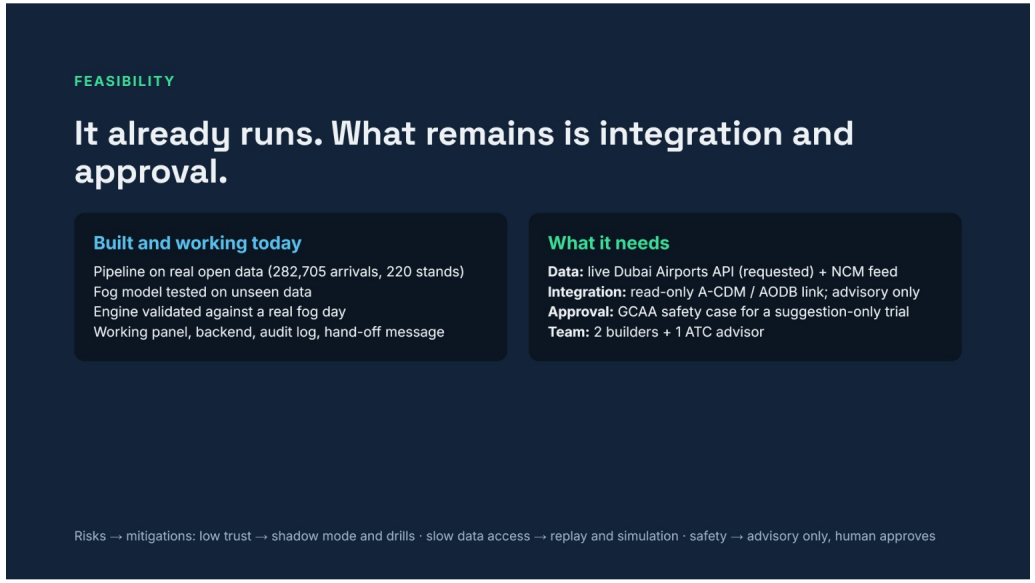
In our replay of the real 3 January morning, DABAB's plan used 37% less holding and diversion fuel than an uncoordinated plan in the same simulation, and cut risk by a quarter. The flights, weather and stands are real data; the simulation reproduced a real fog day, 10 February, almost exactly. So we present these as indicative results, and the 90-day plan is how we prove them with controllers. If a judge asks "are these real?": the weather, flights and stands are real government data; the fog model was tested on years it never saw; the plan savings come from a simulation that reproduced a real fog day almost exactly.



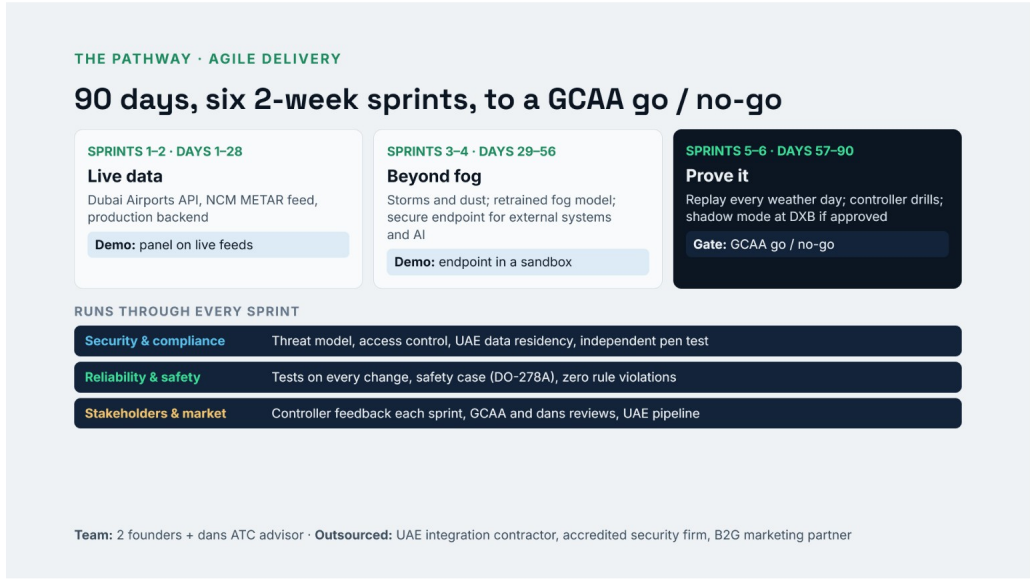
Here is the full model; the next slide has the three points that matter.



The air navigation provider pays an annual licence per airport, with airports as co-buyers. We start at DXB and scale to every UAE airport with the same engine. In our replay, one fog morning saved about 123 tonnes of fuel; we will confirm real savings in the trial.



Most of the hard part already works on real data. What remains is data access, read-only integration and GCAA approval, and DABAB only advises, so nothing in today’s systems has to change.



We deliver in six two-week Agile sprints, each ending in a demo to GCAA and the controllers. First live data, then beyond fog with a secure endpoint for other systems, then proof: every past weather day, drills with controllers and shadow mode if GCAA approves. Security, compliance and safety assurance are not a phase: they run through every sprint. Two of us lead it with an ATC advisor; integration, the security audit and GCC marketing are outsourced to specialist partners. Targets: fog model warns at least 90% of events two or more hours ahead on new data; at least 70% of plans accepted in controller drills; zero safety-rule violations.



Fog will come back to Dubai. With DABAB Copilot, the next time, the plan is ready before the fog is. Thank you.